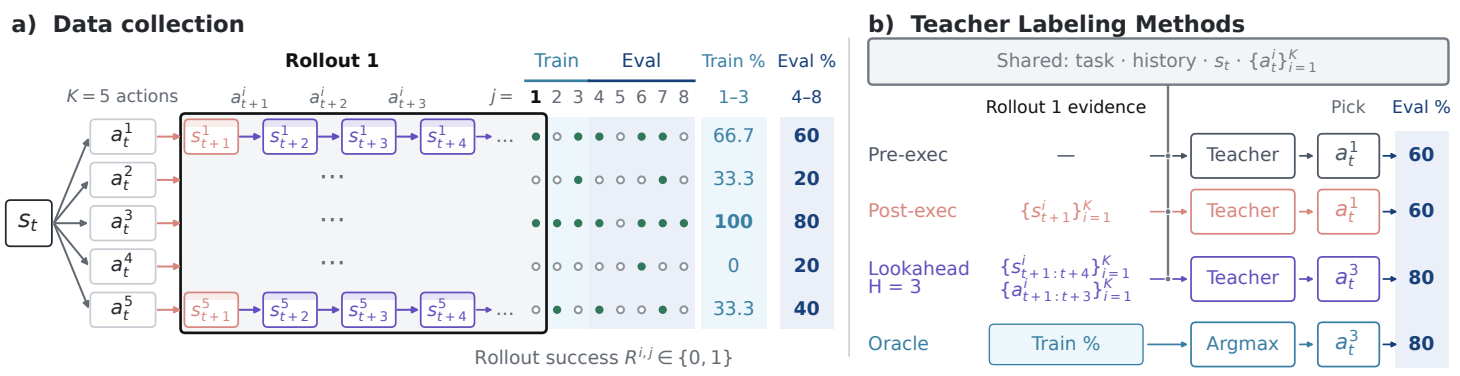
a) Data collection
Rollout 1
Train
Eval
Train %
Eval %
$K = 5$ actions
a^i_{t+1}
a^i_{t+2}
a^i_{t+3}
$j =$ 1 2 3 4 5 6 7 8
1–3
4–8
s_t
a^1_t
a^2_t
a^3_t
a^4_t
a^5_t
s^1_{t+1} s^1_{t+2} s^1_{t+3} s^1_{t+4} …
s^5_{t+1} s^5_{t+2} s^5_{t+3} s^5_{t+4} …
66.7 60
33.3 20
100 80
0 20
33.3 40
Rollout success $R^{i,j} \in \{0, 1\}$
b) Teacher Labeling Methods
Shared: task · history · s_t · $\{a^i_t\}^K_{i=1}$
Rollout 1 evidence
Pick
Eval %
Pre-exec
—
Teacher
a^1_t
60
Post-exec
$\{s^i_{t+1}\}^K_{i=1}$
Teacher
a^1_t
60
Lookahead
H = 3
$\{s^i_{t+1:t+4}\}^K_{i=1}$
$\{a^i_{t+1:t+3}\}^K_{i=1}$
Teacher
a^3_t
80
Oracle
Train %
Argmax
a^3_t
80